



Smart Recipe Discovery System Using Ingredients & Keywords



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Abstract

In today's fast-paced lifestyle, deciding what to cook using the ingredients available at home has become a common challenge. Most existing recipe platforms require users to search by exact dish names, which is often inconvenient and time-consuming. This limitation leads to inefficient use of ingredients, increased food wastage, and difficulty in making quick cooking decisions.

The Smart Recipe Discovery System addresses this problem by enabling users to search for recipes based on available ingredients or simple keywords. The system processes user inputs to generate relevant and filtered recipe suggestions, along with options based on cuisine, preparation time, and dietary preferences. By simplifying the search process and improving accessibility, the system helps users save time, reduce wastage, and make smarter everyday cooking choices.



Problem Statement



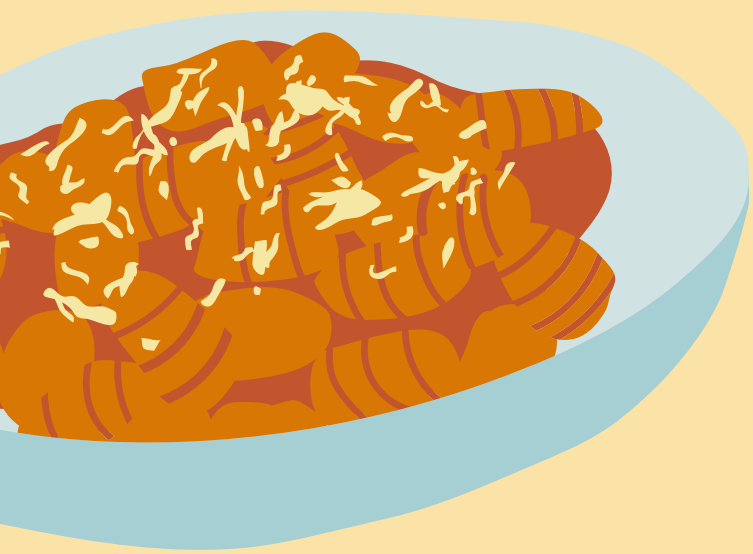
Many users face difficulty in identifying suitable recipes using the ingredients available at home. Existing recipe platforms primarily rely on searching by dish names, making it inconvenient for users who are unsure of what to cook.

This leads to:

- Inefficient use of available ingredients
- Increased food wastage
- Time-consuming browsing and decision-making
- Lack of personalized recommendations

There is a need for an intelligent system that simplifies recipe discovery based on real-time user needs.





Proposed Functionalities

The system is designed with user-centric and intelligent features:

- Ingredient-based search to find recipes instantly
- Keyword-based search for flexible discovery
- Smart filtering (cuisine, preparation time, difficulty level, diet type)
- Auto-suggestions for ingredients and recipes
- Personalized recommendations based on user preferences
- Bookmark/save favorite recipes for future use
- Step-by-step cooking instructions for better usability.





Tools and Technologies to be used



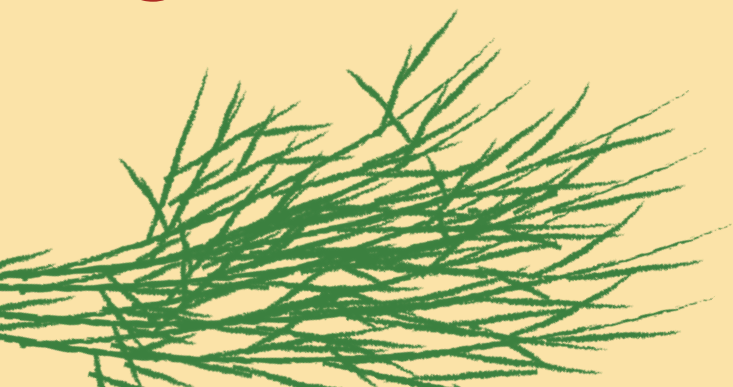
- 1 Frontend: HTML, CSS, JavaScript (for designing an interactive and user-friendly interface)
- 2 Backend: Java (for handling server-side logic and processing user requests)
- 3 Database: MySQL (for storing recipes, ingredients, and user data)

Expected Outcome

The final system will deliver a smart and user-friendly platform that:

- Provides instant recipe suggestions based on available ingredients
- Reduces food wastage through better utilization of resources
- Saves time and effort in decision-making
- Enhances user experience with personalized and filtered results
- Encourages efficient and creative cooking practices

The project ultimately aims to create a practical, scalable, and intelligent solution that improves everyday cooking experiences and aligns with modern user needs.





THANK
YOU!

